

RefCap: Zero-shot Video Corpus Moment Retrieval Based on Refined Dense Video Captioning

Yi Pan^{1,2} Yujia Zhang^{1,*} Michael Kampffmeyer³ Xiaoguang Zhao¹

¹ State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences

² School of Artificial Intelligence, University of Chinese Academy of Sciences

³ Department of Physics and Technology, UiT The Arctic University of Norway

{panyi2022,zhangyujia2014,xiaoguang.zhao}@ia.ac.cn, michael.c.kampffmeyer@uit.no

Abstract—Video corpus moment retrieval (VCMR) is a challenging task aimed at localizing specific segments from untrimmed videos within a vast video collection. It has long been addressed using end-to-end supervised or weakly-supervised methods, which often lack explainability and rely on laborious annotations. To address these issues, we exploit generated captions from Vision Large Language Models (VLLMs) and propose the first zero-shot training-free VCMR system, RefCap. The system consists of two decoupled stages: the Construction Stage to propose dense events and construct corresponding captions, and the Retrieval Stage to retrieve events based on text similarities between queries and captions. In the Construction Stage, we design a Sliding-Window Denoiser and a Quality-Measured Event Generator to produce high-quality dense captions, and derive the Indexing Keyword Sets to fully utilize the generated captions. In the Retrieval Stage, we propose a multi-granularity retrieval strategy integrating both sentence-level and word-level textual similarities between queries and event captions. Extensive experiments on the Charades and ActivityNet datasets demonstrate the effectiveness and competitiveness of our method. Code is available at <https://github.com/BUAAPY/RefCap>.

Index Terms—Video Corpus Moment Retrieval, Dense Video Captioning, Vision Large Language Models

I. INTRODUCTION

Localizing specific segments from untrimmed videos within a vast video collection (i.e. Video Corpus Moment Retrieval, VCMR) is a challenging yet promising task for video retrieval systems. While most previous methods have addressed this task through supervised [1–3] or weakly-supervised [4, 5] training, these methods rely on datasets with laborious annotations. Furthermore, existing methods compute video-text similarities based on implicit video embeddings, which fails to provide explicit semantics of videos and lacks explainability.

To tackle these issues, a promising way is to utilize automatically generated frame captions from Vision Large Language Models (VLLMs) [6, 7]. Frame captions are explicit and interpretable but also contain noise [8] which may mislead models in the process of video understanding [8, 9]. Consequently, several recent works in other video understanding domains have utilized generated captions indirectly, such as employing them as pseudo-annotations [8] or auxiliary data for feature augmentation [9]. Although these methods partially address the noise issue, they are not explicitly designed to do so

and their potential to leverage the generated captions is thus limited.

In this work, we propose the first zero-shot training-free VCMR system, **RefCap**, which directly leverages frame captions generated from VLLMs. Unlike previous works [1–3], RefCap consists of two decoupled stages, which generate and then utilize explicit and readable video captions to achieve both improved accuracy and explainability: The first stage, the Construction Stage, proposes dense events and constructs **Refined Captions** for each video. These are then leveraged in the Retrieval Stage, which retrieves events based on multi-granularity textual similarities between queries and the obtained captions. The Construction Stage is conducted only once and includes two modules to ensure the quality of generated proposals and captions: a Sliding-Window Denoiser (SWi-Den) that reduces low-quality captions by replacing them with neighboring high-quality captions within a sliding window, and a Quality-Measured Event Generator (QM-Gen) based on text similarities that ensures high-quality events by taking into account the quality of captions. To fully utilize frame captions, we further extract Indexing Keyword Sets for each event and propose a multi-granularity retrieval method considering both sentence and word similarities in the Retrieval Stage. Without training on any annotation, RefCap achieves competitive performance compared to weakly-supervised methods on the Charades [10] dataset, and outperforms these methods with a notable margin on the challenging ActivityNet [11] dataset.

Our contributions can be summarized as four folds: (1) We propose the first zero-shot VCMR system, RefCap, composed of a decoupled Construction Stage and Retrieval Stage, functioning in a training-free manner. (2) We design two novel denoising modules, i.e. SWi-Den and QM-Gen, to ensure the quality of the proposed events and the corresponding captions. (3) We propose a multi-granularity retrieval method with introduced Indexing Keyword Sets to fully leverage generated captions. (4) Extensive experiments demonstrate the effectiveness and competitiveness of RefCap compared to weakly-supervised methods on two public datasets.

*Corresponding author.

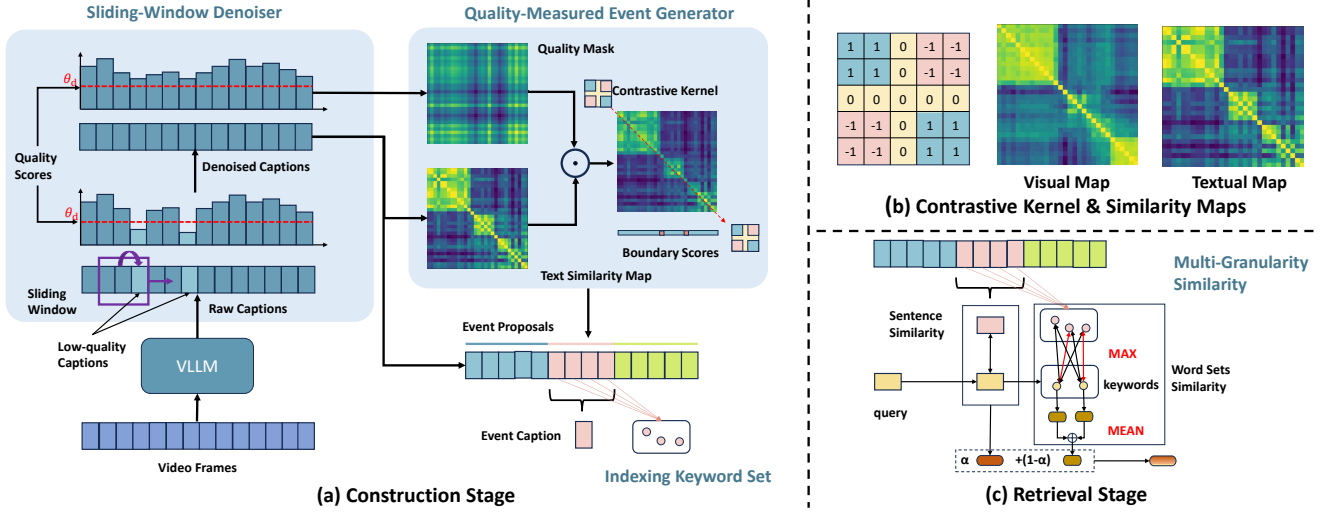


Fig. 1: System overview. (a) In the Construction Stage, raw captions are generated from a Vision Large Language Model and processed through SWi-Den and QM-Gen for dense video captioning. Indexing keyword sets are extracted for each event. (b) The contrastive kernel and the comparison between the visual and text similarity map. The text similarity map contains clearer event boundaries. (c) In the Retrieval Stage, the Multi-Granularity Similarities are computed. The Max_Mean operation is adopted to compute the word sets similarities.

II. METHOD

A. Notations and System Overview

The framework of RefCap is illustrated in Fig. 1. In the Construction Stage, given a video collection with N videos: $\mathbb{V} = \{\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_N\}$, we leverage a Vision Large Language Model to generate raw frame captions $\mathbf{R}_i = VLLM(\mathbf{V}_i) = \{r_1^i, \dots, r_{L_i}^i\}$, where L_i is the number of sampled frames in video $\mathbf{V}_i$. The captions are processed through SWi-Den, QM-Gen and the Indexing Keyword Sets extraction, yielding H_i events with associated captions and keywords: $\mathbf{P}_i = \{p_1^i, \dots, p_{H_i}^i\}$, where each proposal $p_h^i = (s_h^i, e_h^i, c_h^i, K_h^i)$ comprises the start time stamp s_h^i , end time stamp e_h^i , caption c_h^i and Indexing Keyword Set K_h^i .

In the Retrieval Stage, given a text query T , we compute the multi-granularity similarities between T and each event proposal p_h^i , incorporating both sentence-level and word-level similarities. These similarities are leveraged for event-level and video-level retrieval tasks.

B. Dense Video Caption Construction

1) *Sliding-Window Denoiser (SWi-Den)*: For each video $\mathbf{V}_i$, we employ the BLIP [6] model to calculate the image-text matching score between each caption r_l^i and its corresponding frame image f_l^i , and regard the score as the Quality Score of caption r_l^i :

$$Q(r_l^i) = \langle E_v(f_l^i), E_t(r_l^i) \rangle \in \mathbb{R} \quad (1)$$

where E_v and E_t are the visual and text encoders of the BLIP model, respectively. $\langle \cdot, \cdot \rangle$ denotes the inner product operation.

A caption r_l^i is considered as low-quality if its quality score $Q(r_l^i)$ is below the denoising threshold θ_d . Drawing on the insight that adjacent frames share similar semantic content,

we propose a straightforward yet effective method to diminish noise in the raw captions. This is achieved by substituting low-quality captions with nearby high-quality ones. Taking inspiration from traditional filtering techniques, we utilize a sliding window approach to perform the denoising process sequentially from the beginning to the end:

$$r_l^i \leftarrow r_{j^*}^i, Q(r_l^i) < \theta_d \\ j^* = \arg \max_j \{0 < (l - j) \leq W \wedge Q(r_j^i) \geq \theta_d\} \quad (2)$$

where W is the size of the sliding window. The denoised caption set is denoted as $\hat{\mathbf{R}}_i = \{\hat{r}_1^i, \dots, \hat{r}_{L_i}^i\}$.

2) *Quality Measured Event Generator (QM-Gen)*: To ensure high quality event captions, inspired by [12] we generate events based on the frame similarity matrix, where frames from the same event produce high similarity scores. However, different from [12], we utilize text features instead of visual features to compute the similarity matrix M_s^i , since the text features can represent stable high-level semantics and provide clearer event boundaries (see Fig. 1(b)):

$$M_s^i = E_t(\hat{\mathbf{R}}_i) E_t^T(\hat{\mathbf{R}}_i) \in \mathbb{R}^{L_i \times L_i} \quad (3)$$

where $E_t(\hat{\mathbf{R}}_i) \in \mathbb{R}^{L_i}$ are the text features of $\hat{\mathbf{R}}_i$ computed with the BLIP model. To incorporate caption qualities during proposal generation, we introduce the Quality-Mask M_q^i and apply it to mask the text-similarity matrix M_s^i :

$$M_q^i = Q(\hat{\mathbf{R}}_i) Q^T(\hat{\mathbf{R}}_i) \in \mathbb{R}^{L_i \times L_i} \\ \tilde{M}_s^i = M_s^i \odot M_q^i \in \mathbb{R}^{L_i \times L_i} \quad (4)$$

where $Q(\hat{\mathbf{R}}_i) \in \mathbb{R}^{L_i}$ is the Quality Scores for the denoised caption set $\hat{\mathbf{R}}_i$ and $\odot$ is the Hadamard product operation. The

contrastive kernel [12] (Fig. 1(b)) is applied to compute boundary scores along the diagonal of $\widehat{M}_s^i$. Frames with boundary scores exceeding the threshold θ_b are selected as event boundaries, resulting in H_i event proposals $p_h^i = (s_h^i, e_h^i, c_h^i)$, where the frame caption $\hat{r}_{l_h^*}^i$ with the highest Quality Score within each event is chosen as the event caption c_h^i .

3) *Indexing Keyword Set*: To efficiently utilize all generated frame captions while maintaining effectiveness, we employ Spacy [13] to extract keywords (nouns and verbs) from captions in each proposal p_h^i . We take the union set (Indexing Keyword Set) of them for each event to control the scale of the set:

$$K_h^i = \bigcup_{s_h^i \leq l \leq e_h^i} \text{NounVerb}(\hat{r}_l^i) \quad (5)$$

where *NounVerb* denotes the process of extracting nouns and verbs from caption $\hat{r}_l^i$.

With this, all necessary elements for the Retrieval Stage are obtained during the Construction Stage, including proposed events along with their captions and Indexing Keyword Sets for all videos: $\mathbf{P}_i = \{p_1^i, \dots, p_{H_i}^i\}$, $p_h^i = (s_h^i, e_h^i, c_h^i, K_h^i)$. The aforementioned process structures videos at the event-level and is performed only once, ensuring the efficiency of the Retrieval Stage.

C. Multi-granularity Similarity Based Retrieval

Given a text query T , we use the Sentence Transformer [14] (denoted as E_s) to compute the sentence level similarity between T and each event caption c_h^i :

$$S_s(T, c_h^i) = \langle E_s(T), E_s(c_h^i) \rangle \quad (6)$$

For word-level similarity, we extract keywords (nouns and verbs) from T : $K_T = \{k_l^T\}$, and use the Glove [15] model to embed word-level features. The word-level similarity between $k_l^T \in K_T$ and $k \in K_h^i$ is computed as:

$$S_w(k_l^T, k) = \langle E_{glove}(k_l^T), E_{glove}(k) \rangle \quad (7)$$

To measure the similarity between two keyword sets K_T and K_h^i , we define the *Max_Mean* operation as:

$$S_w(T, K_h^i) = \frac{1}{|K_T|} \sum_{k_l^T \in K_T} \max_{k \in K_h^i} \{S_w(k_l^T, k)\} \quad (8)$$

where the inside *max* follows the Multiple Instance Learning [16] paradigm, and the outer average operation measures the matching degree of K_h^i to the overall query T .

1) *Event-level Retrieval*: Event-level retrieval is conducted by selecting events with the highest multi-granularity similarities balanced with hyperparameter α :

$$S(T, p_h^i) = \alpha S_s(T, c_h^i) + (1 - \alpha) S_w(T, K_h^i) \quad (9)$$

2) *Video-level Retrieval*: For partially relevant video-level retrieval, we merge keyword sets $K_i = \bigcup_{1 \leq h \leq H_i} K_h^i$ for each video $\mathbf{V}_i$, and select videos with the highest multi-granularity similarities defined as:

$$S(T, \mathbf{V}_i) = \alpha \max_{1 \leq h \leq H_i} \{S_s(T, c_h^i)\} + (1 - \alpha) S_w(T, K_i) \quad (10)$$

III. EXPERIMENTS

A. Implementation Details

1) *Datasets and Evaluation Metrics*: We conduct experiments on two datasets: (1) **Charades-STA** [10], a dataset with 6672 videos of indoor activities for video moment localization. We use 3720 moment-sentence pairs from the test split. (2) **ActivityNet-Captions** [11], consisting of 20K YouTube videos for dense video captioning and video moment localization. Following [2, 20], we use 17505 moment-sentence pairs from the val-1 split, and adopt Recall@K, IoU=m as the metrics for the Event-Level retrieval subtask, and Recall@K for the Video-Level retrieval subtask.

2) *Experimental Settings*: We employ Blip [6] and MiniGPT [7] as the VLLMs for caption generation. The denoising threshold θ_d is set to 0.4. Frames are uniformly sampled per second, and the window size W is set to 2s. The threshold θ_b in QM-Gen is set to 0.2. The balancing parameter α in (9) and (10) is set to 0.5.

B. Comparison to Weakly-Supervised Methods

We compare RefCap with weakly-supervised methods, and the results are shown in Table I and Table II. The performances of these methods are from JSG [5], the current state-of-the-art weakly supervised method. The results show that RefCap, despite not requiring specific annotations, achieves comparable performances to the best weakly-supervised method JSG on the Charades dataset, and outperforms all the weakly-supervised methods with a large margin on the ActivityNet dataset, which contains longer videos and more complicated text queries. The system using MiniGPT performs better than using Blip, as captions generated by MiniGPT are longer and contain more semantic details.

C. Ablation Study

We conduct ablation studies on both the Construction and Retrieval Stages (Table III). In the Construction Stage, we compare the full setting to systems using (a) visual similarities only, (b) text similarities only, (c) text similarities with SwiDen only, (d) text similarities with QM-Gen only, while maintaining the full setting in the Retrieval Stage. For the Retrieval Stage, we compare the full setting to retrieval strategies using: (e) only sentence-level similarities, (f) only word-level similarities, and (g) joint similarities with *Max_Max* operation (replacing the average operation in (8) with max operation), with the Construction Stage using the full setting. Comparison between experiments (a) and (b) demonstrates the superiority of utilizing text similarities over visual similarities

TABLE I: Event-level performance comparison between RefCap (Zero-shot, ‘B’ for method using Blip model, ‘M’ for method using Minigtpt model) and weakly-supervised methods on Charades and ActivityNet datasets. **Bold**, underline, wave denote the 1st, 2nd, 3rd best results, respectively (same for below).

Method	Charades						ActivityNet					
	Iou=0.3		Iou=0.5		Iou=0.7		Iou=0.3		Iou=0.5		Iou=0.7	
	R@10	R@100	R@10	R@100	R@10	R@100	R@10	R@100	R@10	R@100	R@10	R@100
MCN[17]	-	-	0.52	2.96	0.31	1.75	-	-	0.18	1.26	0.09	0.70
CAL [18]	-	-	0.75	4.39	0.42	2.78	-	-	0.21	1.58	0.10	0.90
XML [19]	0.70	2.47	0.32	1.42	0.16	0.78	3.21	12.48	1.69	7.58	0.10	0.90
HMAN [1]	-	-	1.40	7.79	1.05	4.69	-	-	0.66	4.75	0.32	2.27
MS-SL [20]	4.46	17.61	2.55	10.05	0.91	3.76	10.80	28.31	5.85	15.65	2.46	6.60
JSG [5]	7.66	31.10	6.56	24.76	3.01	13.04	<u>11.36</u>	<u>38.08</u>	<u>5.81</u>	<u>20.54</u>	2.54	8.96
RefCap(B)	<u>5.38</u>	<u>22.37</u>	3.95	<u>15.40</u>	2.18	<u>7.80</u>	<u>16.51</u>	<u>39.65</u>	10.23	<u>24.32</u>	<u>5.00</u>	<u>11.36</u>
RefCap(M)	<u>6.37</u>	<u>27.61</u>	<u>4.25</u>	<u>18.31</u>	<u>2.12</u>	<u>8.90</u>	18.88	43.31	11.47	26.80	5.60	12.61

Input Video												
GT	Person closes the door.				Person plays on his phone.				Person fixes the doorknob.			
w/o Swi-Den & QM-Gen	A door with a lamp and a mirror in the room		A man in a plaid shirt stands in front of a door holding a cell phone				A man in a striped shirt is standing in front of a mirror		A man in a plaid shirt is holding a door handle and adjusting it.			
Full Setting	A door with a lamp and a mirror in the room		A man in a plaid shirt stands in front of an open door.		A man in a plaid shirt stands in front of a door holding a cell phone			A man in a striped shirt is standing in front of a mirror		A man in a plaid shirt is holding a door handle and adjusting it.		

Fig. 2: Visualization of generated dense video captions. Compared to ground truth moments and queries, RefCap in the full setting is able to generate high-quality moments with explicit video captions.

TABLE II: Performances on the Partially Relevant Video Retrieval task.

Method	Charades			ActivityNet		
	R@1	R@5	R@10	R@1	R@5	R@10
XML [19]	1.6	6.0	10.1	5.3	19.4	30.6
DE++ [21]	<u>1.7</u>	5.6	9.6	5.3	18.4	29.2
RIVRL [22]	1.6	5.6	9.4	5.2	18.0	28.2
MS-SL [20]	<u>1.8</u>	7.1	11.8	<u>7.1</u>	<u>22.5</u>	<u>34.7</u>
JSG [5]	1.9	7.0	<u>11.7</u>	6.1	20.1	31.6
RefCap(B)	1.6	5.6	9.0	9.8	24.8	<u>34.3</u>
RefCap(M)	<u>1.8</u>	<u>6.8</u>	<u>11.3</u>	12.0	28.5	39.1

TABLE III: Ablation studies on both the construction and retrieval stages on the Charades dataset.

Id	Method	Event-Level		Video-Level	
		IoU=0.5	IoU=0.7	R@10	R@100
a	Vis	13.74	7.02	10.6	42.6
b	only Txt	16.69	8.55	10.9	44.3
c	Txt & Swi-Den	16.96	<u>8.60</u>	11.3	44.5
d	Txt & QM-Gen	<u>17.98</u>	8.01	10.6	<u>44.9</u>
e	Sent	16.18	8.33	9.9	40.7
f	Word-MaxMean	14.09	6.32	7.5	37.2
g	Joint-MaxMax	17.12	8.28	10.8	43.7
h	Full	18.31	8.90	11.3	45.2

in the event proposal process. Systems without QM-Gen (c) or Swi-Den (d) underperform the full-setting, highlighting the limitations of using raw captions directly. Results from experiments (e), (f), (g) indicate the effectiveness of the proposed Indexing Keyword Sets and the multi-granularity

similarities with *Max_Mean* operation.

D. Visualization

We visually compare the generated dense captions with the ground truth (Fig. 2). Different from [2, 5] that can only produce video-text similarities using implicit video embeddings, RefCap is capable of generating readable dense captions and thus improves explainability.

IV. CONCLUSION

We propose RefCap, the first zero-shot, training-free VCMR system composed of decoupled Construction and Retrieval Stages. In the Construction Stage, we leverage Vision Large Language Models to generate explicit frame captions for given videos, then a sliding-window based denoiser Swi-Den and a quality-measured proposal generator QM-Gen are designed to generate refined dense event captions. We further extract Indexing Keyword Sets to fully utilize the generated captions. In the Retrieval Stage, a Multi-Granularity Similarity Based Retrieval strategy is proposed to retrieve target moments based on pure text similarities. Experiments comparing RefCap to weakly-supervised methods on two public datasets demonstrate its effectiveness and competitiveness.

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